Detecting Weak Structure in Stochastic Particle Systems: A Matched-Counterfactual Multi-Collector Benchmark

Anonymous Authors

Abstract

We develop a matched-counterfactual instrument for testing whether mobile collectors exploit a weak latent transport field beyond capture induced by passive advection. Each signal episode is paired with a no-signal episode sharing its initial state, Brownian forcing, field nuisance variables, and policy randomness. Fixed-geometry contact is solved along exact piecewise specular paths, and exact ownership ties are event-keyed. A 12-seed exploratory calibration produced no positive simultaneous lower bound, while stationary and random controls descriptively matched or exceeded the local-flow policy. We therefore corrected the first-event estimand to subtract the stationary signal–null response and introduced a catchability ratio that separates drift strength from collector speed. A distinct full-state action-feasible oracle failed that preregistered first-event feasibility gate after one bounded state-input repair: first contacts occurred within a few steps, indicating endpoint saturation rather than a compute shortage. We then changed only the endpoint and stopping rule to count unique team captures through the inclusive end of step 67, one sensing-radius traversal at maximum collector speed. On eight fresh diagnostic seeds at $\alpha = 0.06$, the oracle exceeded stationary collectors by 9.375 particles on average (8/8 positive) and the true-field-only control by 7.5 (7/8 positive); all frozen correctness, direction, targeting, pairing, checksum, and artifact gates passed. A separately frozen eight-seed coupled-noise diagnostic then found mean oracle-minus-stationary yield contrasts of 8.625 particles at both $\Delta t = 0.02$ and 0.01, with zero seed-level direction changes; the mandatory $\Delta t = 0.005$ result was 8.375. All preregistered numerical gate components passed. These results establish diagnostic action-contingent headroom and timestep robustness for the redesigned endpoint only. They authorize, but do not answer, the bounded shared-summary diagnostic. That treatment remains unexecuted and SPS-C03 remains blocked. **No coordination effect, power, confirmation, learning, or MARL result is claimed.**

1 Related work

Stochastic particle capture and transport. Particle capture in flowing media spans a rich theoretical and experimental literature. Maxey [3] characterized inertial-particle settling in turbulent flows and identified the mechanisms by which particles cluster relative to fluid trajectories. Sommerer and Ott [4] showed that passive tracers in chaotic two-dimensional flows develop striking spatial structure whose statistics depend sensitively on the Lyapunov exponent, establishing the dynamical-systems framework that underlies chaotic-flow benchmarks. First-passage theory [5] supplies the mathematical scaffold for capture-time estimands: the expected time to first contact between a diffusing target and an absorbing boundary depends on both geometry and drift in ways that motivate our matched signal–null framing. Wang et al. [1] demonstrated that a stationary collector can exploit chaotic-flow structure to improve capture rates and characterised placement and capacity effects. Wang et al. [2] extended this to a mobile collector that observes its local particle distribution and selects from four motion strategies, reporting scaling laws for capture time. Their conclusion explicitly names multiple collectors and coordination as future work. We do not claim the first mobile particle collector, first locally guided capture strategy, or first capture-time analysis. The distinguishing element is a matched counterfactual design that pairs each signal episode with a null episode sharing initial state, Brownian forcing, and randomness, so that the seed-level yield difference is a matched policy contrast conditional on identical recorded stochastic inputs. Stationary and true-field controls are reported separately because matching alone does not remove passive advection as an explanation.

Multi-agent reinforcement learning and coordination. MADDPG [6] established centralized-training decentralized-execution actor-critic learning for continuous mixed cooperative–competitive environments. QMIX [8] introduced monotonic value-function factorisation that scales cooperative credit assignment to larger teams. MAPPO and IPPO [7] subsequently showed that independent proximal-policy optimization matches or exceeds joint-training baselines across several cooperative benchmarks when implemented carefully. On the communication side, DIAL [9] and IMAC [10] demonstrated that agents can learn compact, bandwidth-constrained message protocols jointly with behavior policies. These methods learn communication jointly with action selection; the protocols that emerge are difficult to interpret causally as isolated mechanisms. Our approach is orthogonal and prior to learning: we ask whether a single fixed, hand-coded team-mean velocity summary shifts the capture yield boundary compared with four identically capable independent collectors, at the same action budget and under matched randomness. This is a mechanism probe that precedes any learning experiment. It cannot demonstrate that an optimal multi-agent policy exists, but it can rule out or confirm that a specific low-dimensional shared statistic provides actionable information beyond independent parallel search. We therefore do not claim the first multi-agent coordination result, the first communication protocol, or the first cooperative MARL method.

Active particles and collective foraging. The physics of self-propelled particles constitutes a mature subfield of active matter [12]. Vicsek et al. [13] established that simple local alignment rules among self-propelled particles produce collective ordered motion through a phase transition, a foundational result for multi-agent collective behavior. Berdahl et al. [14] provided experimental evidence that animal groups collectively sense complex environmental gradients—in particular, that individual agents responding to local cues can produce group-level gradient ascent even when no individual can detect the gradient—directly motivating the question of whether shared local summaries shift a collective exploitation boundary. On the algorithmic side, Löffler et al. [11] trained thirty locally observing active colloids with a shared PPO policy to forage for randomly appearing food, finding that collective flocking and milling emerge without explicit coordination rewards. We do not claim the first collective foraging system, first active-matter multi-agent benchmark, first local-sensing RL particle task, or first communication-enabled collective sensing paper. The role reversal relative to Löffler et al. is precise: in that work the particles are controlled agents and food is directly perceived; here a small collector team acts while passive particles are environment entities and the transport signal must be inferred from particle motion histories.

Counterfactual and causal inference in multi-agent systems. COMA [15] introduced counterfactual baselines into multi-agent policy gradients to isolate each agent's marginal contribution by comparing actual returns with expected returns under a default action, holding other agents' actions fixed. The matched episode design in this work is structurally analogous: seed-level outcomes under the shared and independent policies are compared on identical stochastic inputs, producing a paired potential-outcome contrast under the recorded simulator intervention [16, 17]. The three candidate contributions, conditional on the pending numerical and shared-versus-independent diagnostics, are: (i) a matched counterfactual causal framing for multi-collector particle capture that subtracts the passive-advection response, so the estimand measures action-contingent exploitation rather than passive transport; (ii) exact piecewise-specular contact geometry with event-keyed tie-breaking, so captured-particle identity is deterministic given the matched random inputs and does not depend on entity-array ordering; and (iii) a bounded-statistic mechanism probe that precedes any learning experiment, testing whether one specific hand-coded three-number summary shifts the yield boundary before introducing the confounds of a learned communication protocol. If the diagnostic fails, only the simulator and negative mechanism audit remain; no coordination contribution is claimed. If it passes, these properties make SPS-C03 a mechanism diagnostic rather than a coordination-algorithm contribution; the result, whether positive or null, is interpretable without reference to a particular learning algorithm.

2 Research question and novelty boundary

The intended paper asks one question: *with four collectors in the frozen unit-square task at $\alpha = 0.06$, does replacing each collector's independent local velocity estimate with the same bounded team-mean estimate increase the number of unique particles captured through the inclusive end of step 67?* This comparison is presently blocked rather than answered: the redesigned endpoint has passed its upstream action-sensitivity gate, but coupled-noise timestep validation and a separate sharing preregistration must precede any shared-versus-independent execution.

Prior work already studies stationary capture in chaotic flow [1], a locally informed mobile collector capturing passive particles [2], local-communication collective sensing [18], learned and bandwidth-aware multi-agent communication [9, 10], and local-sensing reinforcement learning for active-particle foraging [11]. We therefore do not claim the first mobile particle collector, locally guided capture task, distributed flow-sensing framework, bounded communication method, particle-foraging problem, or local-sensing learning formulation. A fixed team mean is only a mechanism probe. The defensible target, if upstream gates eventually pass, is the causal effect of sharing that precisely defined statistic under matched sensing, action, and policy-capacity budgets, beyond passive advection and independent parallel search. Exact within-step intersection is likewise not a numerical novelty or a substitute for convergence of stochastic hitting statistics [19, 20].

3 Frozen system and local policy

The canonical domain is the unit square with reflecting boundaries. Four collectors act among $N = 256$ non-learning particles for $H = 400$ steps with $\Delta t = 0.02$ and $\sigma = 0.06$. Before capture, particle k follows

$$x_{k,t+1} = \mathcal{R}\Big(x_{k,t} + \Delta t\, \alpha b_z(x_{k,t}) + \sigma\sqrt{\Delta t}\, \epsilon_{k,t}\Big), \qquad \epsilon_{k,t} \sim \mathcal{N}(0, I_2),$$

where the primary field is spatially uniform. Collector action $a_{i,t} \in \mathbb{R}^2$ is projected onto the unit ball before a common speed limit $v_{\max} = 0.12$ is applied:

$$y_{i,t+1} = \mathcal{R}\left(y_{i,t} + \Delta t\, v_{\max} \frac{a_{i,t}}{\max\{1, \|a_{i,t}\|_2\}}\right).$$

Collectors reset in ID order at $(.25, .25)$, $(.25, .75)$, $(.75, .25)$, $(.75, .75)$. Particles are rejection-sampled from a dedicated initialization stream until strictly outside every initial capture disc. The episode pre-generates its full Brownian tensor and draws one field orientation from a separate stream.

The frozen `local_flow_v1` rule acts independently for each collector. It averages only causally valid visible-particle velocity slots, moves at unit normalized speed opposite that mean, and stops when none is valid. It has no density fallback and ignores the available teammate-position channel. Thus the nominal four-agent treatment is exactly four independent replicas, not a coordination treatment.

Each local observation contains the arena-normalized self position; up to $K = 32$ nearest free particles within radius 0.16; masks; and teammate positions relative to self in fixed ID order. A particle velocity is emitted only when that particle was visible to the same collector in both consecutive observations. The interface receives no field parameter, global state, future noise, or random generator.

4 Exact fixed-geometry contact and event-keyed ties

Endpoint contact is inadequate when either body reflects during a step. The implemented fixed-geometry rule retains each unfolded proposal, finds the union of particle and collector wall-hit fractions, and decomposes the step into piecewise-linear rectangular specular segments. On each interval, the earliest disc–point contact is the first admissible root of the relative-motion quadratic. Ownership is ordered by earliest contact fraction, then closest segment distance.

Any remaining exact tie is sampled from sorted stable collector IDs using `SeedSequence`(s, t, k), where s is the scenario seed, t the one-based step, and k the stable particle ID. The decision therefore does not depend on previous random-number consumption, entity-array order, or unrelated events. The solver is exact for the piecewise-linear specular path inside one Euler step; it is not evidence that the discretized stochastic dynamics have converged as $\Delta t \to 0$. Growing aggregates retain deferred endpoint semantics and are excluded from the primary experiment.

5 First-event diagnostic history, physical axes, and provenance

Let $T^*_{s,\pi,c} \in \{1, \ldots, H, H+1\}$ be the one-based first-interception step for scenario seed s, policy π, and condition c, with $H+1$ denoting no contact. The policy-specific signal–null gain is

$$G_s(\pi, \alpha) = \frac{T^*_{s,\pi,0} - T^*_{s,\pi,\alpha}}{H}.$$

Because the same signal can change passive flux, the corrected mechanism estimand subtracts the response of stationary collectors:

$$A_s(\pi, \alpha) = G_s(\pi, \alpha) - G_s(\text{stationary}, \alpha).$$

For the superseded first-event coordination comparison,

$$\Gamma_s(\alpha) = A_s(\text{shared}, \alpha) - A_s(\text{independent}, \alpha) = G_s(\text{shared}, \alpha) - G_s(\text{independent}, \alpha).$$

The stationary term cancels algebraically in Γ_s, but all four absolute signal/null outcomes and both passive-adjusted effects must still be reported so that passive flux and endpoint saturation remain visible.

The active redesigned endpoint instead lets $Y_s(\pi, \alpha)$ denote the number of unique particles captured by the team through the inclusive end of step 67. The episode continues after its first contact. The window is

$$L = \left\lceil \frac{0.16}{0.12 \times 0.02} \right\rceil = 67,$$

so a maximum-speed collector can traverse one sensing radius. The frozen upstream contrast is $d_s = Y_s(\text{oracle}, 0.06) - Y_s(\text{stationary}, 0.06)$, with practical threshold $E[d_s] \geq 4$, corresponding to one additional capture per collector. A mandatory targeting contrast replaces stationary with the true-field-only upstream control. Any later coordination estimand will be the matched seed-level difference $Y_s(\text{shared}, 0.06) - Y_s(\text{independent}, 0.06)$; its minimum relevant effect, inference procedure, and seed budget have not been frozen.

Two dimensionless coordinates are required:

$$\rho = \frac{\alpha\sqrt{\Delta t}}{\sigma}, \qquad \kappa = \frac{\alpha}{v_{\max}}.$$

Here ρ compares drift with per-step Brownian displacement, whereas κ compares particle drift with collector control authority. Under the canonical $\Delta t = 0.02$, $\sigma = 0.06$, and $v_{\max} = 0.12$, the old $\rho = 2$ point has $\kappa \approx 7.07$; it is therefore a poor catchability diagnostic. The bounded oracle gate instead used $\kappa \in \{0.25, 0.5, 1.0\}$. Complete scenario-seed blocks remain the sampling units. Any future boundary requires a separately preregistered grid, minimum effect, simultaneous lower bound, and independent confirmation seeds.

The matched runner rejects a pair unless its environment configurations differ only in signal strength and the null strength is zero. It verifies identical particle and collector initialization, Brownian tensors, and field nuisance variables. It records SHA-256 digests of those inputs, policy randomness, and the event-key specification. At zero signal, full records must be identical after ignoring only the condition label.

Each pair is schema-validated before an immutable write. The runner produces null and signal JSONL trajectories, a pair summary, and a manifest containing the repository commit and dirty flag, environment-config digest, runtime command and versions, policy privilege flag, and artifact byte counts and SHA-256 digests. The bounded pilots retain compact pair summaries and aggregate analyses; they are not a released trajectory dataset.

6 Studentized simultaneous bootstrap

Rows of effect matrix G are scenario-seed blocks and columns are the five nonzero preregistered values $\rho \in \{0.1, 0.25, 0.5, 1, 2\}$. Each bootstrap draw resamples complete rows, retaining dependence across the grid. With observed means $\bar{G}_j$ and standard errors $\widehat{\mathrm{se}}_j$, the common correction is

$$q_{.95} = Q_{.95}\left[\max_j \frac{\bar{G}_j - \bar{G}^*_j}{\widehat{\mathrm{se}}^*_j}\right], \qquad L_j = \bar{G}_j - q_{.95}\widehat{\mathrm{se}}_j.$$

The frozen implementation uses 10,000 draws and bootstrap seed 73,031.

Synthetic null calibration used 2,000 trials, 2,000 bootstrap draws per trial, 12 rows, and five columns. The familywise false-crossing rate was 0.040 under a correlated continuous null (Wilson 95% upper bound 0.0495) and 0.006 under a zero-inflated symmetric discrete null (upper bound 0.0105). Both met the predeclared acceptance rule that this upper bound not exceed 0.07. This checks only synthetic calibration, not the scientific hypothesis.

7 Twelve-seed exploratory calibration

SPS-P02 used scenario seeds 1001–1012 to reproduce the frozen calibration after the exact contact repair. These seeds were assigned to runtime, variance, censoring, oracle-feasibility, and contact diagnostics. They *cannot* update SPS-C01. Table 1 documents the exploratory calculation.

Table 1: Exploratory `local_flow_v1` calibration. Each row uses the same 12 pilot seeds; LCBs are simultaneous one-sided 95% bounds.

ρ	Mean D	Paired SE	Simultaneous LCB
0.10	0.00563	0.00339	-0.01533
0.25	0.00375	0.00457	-0.02452
0.50	0.00708	0.00423	-0.01908
1.00	0.00896	0.00459	-0.01944
2.00	0.01104	0.00514	-0.02079

The studentized maximum critical value was 6.187. Every LCB was negative, so the exploratory rule returned right-censoring beyond the largest tested point, $\rho = 2$. Because the seeds are calibration-only, this neither supports nor refutes a finite boundary and is not a boundary estimate.

Table 2 shows the strongest-signal controls. Stationary, pre-generated random, and privileged upstream policies descriptively matched or exceeded the four-independent local-flow mean. The one-collector row is not capacity-matched and is diagnostic only. No inferential comparison among these controls was authorized.

Table 2: Exploratory controls at $\rho = 2$, each with 12 calibration seeds.

Policy / treatment	Mean D	Median D
Stationary, $M = 4$	0.01313	0.00625
Pre-generated random, $M = 4$	0.01292	0.00875
Privileged upstream oracle, $M = 4$	0.01292	0.00625
`local_flow_v1`, $M = 4$ independent	0.01104	0.00375
Coverage, $M = 4$	0.00771	0.00625
Density greedy, $M = 4$	0.00625	0.00250
`local_flow_v1`, $M = 1$, diagnostic	0.06938	0.01625

Contact-repair comparison. SPS-P01 guarded every reflected path pair; SPS-P02 used exact specular segments. Across the same 144 matched pairs, guarded diagnostics fell from 28,929 to zero and zero of 144 first-interception outcomes changed. This supports the regression repair, not independent scientific replication.

Power status. The earlier provisional 24-seed plan targeted a half-width of 0.05, which is coarser than the exploratory effects of approximately 0.004–0.011. It is not a scientifically usable confirmation budget and SPS-E01 remains blocked. No simulation-based power study has been run for the passive-adjusted or coordination estimand. **PENDING: After a redesigned endpoint passes its oracle and timestep gates, freeze a minimum relevant effect and calibrate type-I error and power over complete scenario-seed blocks before authorizing confirmation.**

8 First-event oracle failure on the catchable grid

The legacy `privileged_upstream_oracle` is renamed conceptually as a *true-field upstream control*: it knows the true field direction and moves upstream, but it neither observes every target position nor computes a feasible intercept. It is not an interception oracle. The distinct full-state oracle observes all free-particle positions and the known current deterministic drift, assigns distinct feasible targets to collectors over a bounded receding horizon, obeys the same action and speed limits, and receives no future Brownian increments.

SPS-P03 exposed an implementation error: the first oracle version treated the last realized Brownian displacement as predictive motion state. The only permitted state-input repair replaced that quantity with current deterministic drift and was preregistered as SPS-P04. The corrected oracle still failed every nonzero feasibility point (Table 3). This is an exploratory diagnostic gate, not a significance test or confirmatory claim.

Table 3: SPS-P04 corrected full-state-oracle gate on eight diagnostic seeds. Passing required positive mean passive-adjusted gain, at least six positive seed directions, and favorable absolute comparisons. No point passed.

α	κ	Mean A_s	Positive seeds	Gate
0.03	0.25	-0.00844	0/8	Fail
0.06	0.50	-0.00563	1/8	Fail
0.12	1.00	-0.01063	2/8	Fail

The failure is consistent with first-event saturation in the canonical reset. Across the diagnostic grid, stationary collectors averaged 5.78 steps to first contact and the corrected oracle averaged 3.00; every episode was uncensored, so the 400-step horizon never bound. The early contacts showed no wall involvement. The oracle can improve absolute null performance while still having a negative policy-minus-stationary signal response because passive signal transport benefits stationary collectors strongly and leaves little action-contingent headroom in the signal arm. Thus the current first-event endpoint and initial geometry do not identify the proposed mechanism.

Per the preregistered stop rule, SPS-P04 blocks SPS-C03 and all downstream scientific execution. A later run that exceeded the one-repair allowance is preserved for audit but procedurally excluded from every claim, manuscript decision, design choice, and power calculation; none of its outcomes are used here.

9 Fixed-horizon action-sensitivity gate

SPS-WO-05 changed only the endpoint and episode stopping rule. The unit-square arena, canonical collector lattice, 256 particles, four collectors, field, diffusion, capture geometry, actions, policies, and matched random streams were unchanged. Captured particles were counted once by stable particle ID, episodes continued after first contact, step 67 was included, and step 68 was excluded. The frozen diagnostic used fresh scenario seeds 2001–2008, field strengths $\alpha \in \{0, 0.06\}$, and stationary, true-field-only, and full-state-oracle policies. These seeds and outcomes are permanently ineligible for confirmation.

At $\alpha = 0.06$, the oracle-minus-stationary seed contrasts were

$$(11, 13, 9, 12, 7, 7, 11, 5),$$

with mean 9.375, median 10, sample standard deviation 2.825, and positive direction in 8/8 seeds. The descriptive paired-bootstrap 95% interval was $[7.5, 11.25]$. Oracle-minus-true-field-control contrasts were

$$(6, 12, 6, 15, 8, 3, 11, -1),$$

with mean 7.5, median 7, sample standard deviation 5.155, and positive direction in 7/8 seeds; the descriptive paired-bootstrap 95% interval was $[4.125, 10.875]$. All frozen gate components passed: deterministic correctness and execution, mean oracle gain of at least four, positive primary direction in at least six seeds, positive mean targeting contrast, matched streams, checksums, and artifact completeness. The intervals are descriptive, not confirmatory.

The valid package contains all 48 policy-condition episodes and 3,216 environment steps. A process-control race caused an earlier identical process to finish after it had been prematurely classified as invalid. Its episode and event files are bitwise identical to the canonical R1 package; it is retained only for audit and is neither pooled, counted as replication, nor used to improve precision. R1 is the sole evidence package.

This result repairs the upstream feasibility diagnosis: unlike time to first contact, fixed-horizon unique yield leaves measurable room for legal actions to matter. It does not show that local observations are sufficient, that sharing helps, or that any learned multi-agent policy can exploit this headroom.

10 Coupled-noise timestep diagnostic

SPS-WO-06 tested the fixed physical duration 1.34 at $\Delta t \in \{0.02, 0.01, 0.005\}$, corresponding to 67, 134, and 268 steps. For each of fresh diagnostic seeds 3001–3008, Brownian increments were generated at the finest resolution and summed to obtain the coarser paths. Stationary and full-state-oracle episodes therefore used a genuinely coupled refinement rather than unrelated noise draws. All 96 preregistered episodes completed, totaling 15,008 environment steps. These seeds and outcomes are permanently ineligible for confirmation.

At $\alpha = 0.06$, the oracle-minus-stationary seed contrasts were

$$(5, 13, 8, 12, 7, 10, 8, 6)$$

at $\Delta t = 0.02$,

$$(4, 16, 8, 9, 6, 11, 10, 5)$$

at $\Delta t = 0.01$, and

$$(4, 15, 9, 9, 6, 9, 8, 7)$$

at $\Delta t = 0.005$. The corresponding means were 8.625, 8.625, and 8.375 particles, with positive direction in 8/8 seeds at every resolution. The preregistered primary comparison had absolute mean change $0.0 < 1.0$ particle from $\Delta t = 0.02$ to 0.01 and zero of eight seed-level sign changes. The mandatory finest-grid result differed from the $\Delta t = 0.01$ mean by 0.25 particle and was informational rather than a rescue criterion. Correctness, coupling, provenance, and artifact gates all passed.

This is a numerical diagnostic of one policy contrast at one field strength, horizon, geometry, and seed block. It is not a confirmation, a coordination effect, or evidence that the shared summary helps. Passing SPS-WO-06 authorizes only the separately preregistered SPS-WO-07 attribution-control diagnostic; SPS-WO-07 has not been executed.

11 AAMAS relevance failure and blockers

The current treatment is not yet a multi-agent contribution. Teammate positions are observed but unused; the four collectors apply independent copies of one scripted rule; and no learned policy is evaluated. The pilot therefore concerns replicated local control and passive transport, not cooperation, communication, strategic interaction, or MARL. No coordination or MARL claim is supported.

The initial scientific no-go remains relevant to the superseded endpoint. At $\rho = 2$, passive stationary and uninformed random controls descriptively matched or exceeded `local_flow_v1`; at the catchable points, the repaired full-state oracle failed its passive-adjusted first-event gate. The fixed-horizon diagnostic establishes action-contingent headroom, and SPS-WO-06 passes its preregistered coupled-noise timestep gate for the oracle-minus-stationary contrast. Neither result is a multi-agent effect. SPS-C03 therefore remains blocked and the shared-summary policy, although implemented and unit-tested, has not been run as a scientific treatment. SPS-WO-07 is authorized but unexecuted. No empirical power calculation, independent confirmation, learned policy, IPPO/MAPPO baseline, or MARL experiment exists. Growing-aggregate physics and aggregation analysis remain later work.

The dependency-free implementation currently has 96 passing tests covering deterministic reset, causal observations, zero-signal identity, exact specular contacts, event-order and event-key invariance, schema validation, immutable writes, paired provenance, frozen policies, row-order-invariant inference, bounded team-summary invariance, full-state-oracle microcases, and exact Brownian aggregation, as well as unique-yield counting, inclusive step-67 evaluation, post-contact continuation, and matched yield-run provenance.

These are engineering checks only. SPS-C01 was dropped when the first-interception endpoint failed its action-feasible oracle gate; SPS-C03 is blocked, and **no detectability boundary is supported by SPS-P01 or SPS-P02**.

12 Limitations

Any eventual boundary is policy-, horizon-, field-, geometry-, grid-, and resource-relative rather than a universal physical critical point. Common random numbers reduce variance only under causally coherent pairing. Exact within-step specular contact does not by itself establish timestep robustness; SPS-WO-06 supplies only a bounded coupled-noise diagnostic for the fixed-horizon oracle-minus-stationary contrast at $\alpha = 0.06$. The canonical first-contact endpoint remains saturated by initial geometry and passive transport. The fixed-horizon endpoint clears its diagnostic oracle and timestep gates, but AAMAS fit still requires an isolated multi-agent mechanism, not merely four collectors. **PENDING: Run the separately preregistered SPS-WO-07 shared-versus-independent attribution-control diagnostic. Do not proceed to power, learning, or confirmation unless that diagnostic passes its frozen gates.**

References

[1] Wang et al. Particle capture in a model chaotic flow. *Physical Review E*, 104:064203, 2021. https://doi.org/10.1103/PhysRevE.104.064203.

[2] Wang et al. Mobile-collector capture of particles in a chaotic flow. *PLOS ONE*, 20(8):e0329766, 2025. https://doi.org/10.1371/journal.pone.0329766.

[3] M. R. Maxey. The gravitational settling of aerosol particles in homogeneous turbulence and random flow fields. *Journal of Fluid Mechanics*, 174:441–465, 1987. https://doi.org/10.1017/S0022112087000193.

[4] J. C. Sommerer and E. Ott. Particles floating on a moving fluid: A dynamically comprehensible physical fractal. *Science*, 259(5093):334–337, 1993. https://doi.org/10.1126/science.259.5093.334.

[5] S. Redner. *A Guide to First-Passage Processes*. Cambridge University Press, Cambridge, 2001. ISBN 0-521-65248-0.

[6] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. https://arxiv.org/abs/1706.02275.

[7] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu. The surprising effectiveness of PPO in cooperative multi-agent games. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. https://arxiv.org/abs/2103.01955.

[8] T. Rashid, M. Samvelyan, C. Schroeder de Witt, G. Farquhar, J. Foerster, and S. Whiteson. QMIX: Monotonic value function factorisation for deep multi-agent reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018. https://arxiv.org/abs/1803.11605.

[9] J. Foerster, Y. Assael, N. de Freitas, and S. Whiteson. Learning to communicate with deep multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016. https://proceedings.neurips.cc/paper_files/paper/2016/hash/c7635bfd99248a2cdef8249ef7bfbef4-Abstract.html.

[10] R. Wang, X. He, R. Yu, W. Qiu, B. An, and Z. Rabinovich. Learning efficient multi-agent communication: An information bottleneck approach. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020. https://proceedings.mlr.press/v119/wang20i.html.

[11] R. C. Löffler, E. Panizon, and C. Bechinger. Collective foraging of active particles trained by reinforcement learning. *Scientific Reports*, 13:17055, 2023. https://doi.org/10.1038/s41598-023-44268-3.

[12] J. Elgeti, R. G. Winkler, and G. Gompper. Physics of microswimmers—single particle motion and collective dynamics. *Reports on Progress in Physics*, 78(5):056601, 2015. https://doi.org/10.1088/0034-4885/78/5/056601.

[13] T. Vicsek, A. Czirók, E. Ben-Jacob, I. Cohen, and O. Shochet. Novel type of phase transition in a system of self-propelled particles. *Physical Review Letters*, 75(6):1226–1229, 1995. https://doi.org/10.1103/PhysRevLett.75.1226.

[14] A. Berdahl, C. J. Torney, C. C. Ioannou, J. J. Faria, and I. D. Couzin. Emergent sensing of complex environments by mobile animal groups. *Science*, 339(6119):574–576, 2013. https://doi.org/10.1126/science.1225883.

[15] J. Foerster, G. Farquhar, T. Afouras, N. Nardelli, and S. Whiteson. Counterfactual multi-agent policy gradients. In *Proceedings of the 32nd AAAI Conference on Artificial Intelligence (AAAI)*, 2018. https://arxiv.org/abs/1705.08926.

[16] S. L. Morgan and C. Winship. *Counterfactuals and Causal Inference: Methods and Principles for Social Research*, 2nd ed. Cambridge University Press, Cambridge, 2015. https://doi.org/10.1017/CBO9781107587991.

[17] G. W. Imbens and D. B. Rubin. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press, Cambridge, 2015. https://doi.org/10.1017/CBO9781139025751.

[18] H. Hornischer et al. CIMAX: Collective information maximization in robotic swarms using local communication. arXiv:1903.05444, 2019. https://arxiv.org/abs/1903.05444.

[19] E. Gobet. Weak approximation of killed diffusion using Euler schemes. *Stochastic Processes and their Applications*, 87(2):167–197, 2000. https://doi.org/10.1016/S0304-4149(99)00109-X.

[20] M. B. Giles. Multilevel Monte Carlo path simulation. *Operations Research*, 56(3):607–617, 2008. https://doi.org/10.1287/opre.1070.0496.